Composition of Foods
Raw, Processed, Prepared

USDA National Nutrient Database for Standard Reference, Release 27

Documentation and User Guide

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May 2015 Revision

The food items in this revision have been updated in the online web search program as well as in the Microsoft Access database. An update file is available to users who wish to use the ASCII files to import the database into their own applications. The information on the number of records in each file in the SR27 Documentation and Users Guide have not been updated to reflect these revisions. The changes included in this revision will be included in the update files disseminated with SR28.

06164 Sauce, salsa, ready-to-serve
Updated nutrient profile with new data
06931 Sauce, pasta, spaghetti/marinara, ready-to-serve
Updated nutrient profile with new data
11172 Corn, sweet, yellow, canned, whole kernel, drained solids
Updated nutrient profile with new data
14649 Hibiscus tea, brewed from calyces
Updated complete nutrient profile and modified the food description
16051 Beans, white, mature seeds, canned
Updated sodium, vitamin A and several fatty acids
16338 Beans, navy, mature seeds, cooked, boiled, with salt

Updated total dietary fiber, and folate

19042 Snacks, potato chips, barbecue-flavor

Updated nutrient profile with new data

32020 Chicken, thighs, frozen, breaded, reheated

Added values for total dietary fiber

The statistical information associated with the nutrient values for the following food items were recalculated to give better estimates of the variability.

06019	Soup, chicken noodle, canned, condensed
06150	Sauce, barbecue
06159	Soup, tomato, canned, condensed
07028	Ham, sliced, packaged (96% fat free, water added)
11056	Beans, snap, green, canned, regular pack, drained solids
11398	Potato puffs, frozen, unprepared
11531	Tomatoes, red, ripe, canned, packed in tomato juice
11578	Vegetable juice cocktail, canned
16424	Soy sauce made from soy and wheat (shoyu), low sodium
21021	Fast foods, english muffin, with egg, cheese, and canadian bacon
21061	Fast foods, burrito, with beans and cheese
21082	Fast foods, taco with beef, cheese and lettuce, hard shell
21106	Fast foods, fish sandwich, with tartar sauce and cheese
21108	Fast foods, hamburger, single, regular patty, with condiments
21139	Fast foods, potato, mashed
21230	Fast Foods, crispy chicken filet sandwich, with lettuce and mayonnaise
21300	Fast Food, Pizza Chain, 14" pizza, cheese topping, thick crust
21301	Fast Food, Pizza Chain, 14" pizza, cheese topping, thin crust
21302	Fast Food, Pizza Chain, 14" pizza, pepperoni topping, regular crust
21303	Fast Food, Pizza Chain, 14" pizza, pepperoni topping, thick crust
21395	Fast foods, cheeseburger, double, regular patty, with condiments
21486	Fast foods, taco with beef, cheese and lettuce, soft
21487	Fast foods, taco with chicken, lettuce and cheese, soft

Introduction

The USDA National Nutrient Database for Standard Reference (SR) is the major source of food composition data in the United States. It provides the foundation for most food composition databases in the public and private sectors. As information is updated, new versions of the database are released. This version, Release 27 (SR27), contains data on 8,618 food items and up to 150 food components. It replaces SR26 issued in August 2013.

Updated data have been published electronically on the USDA Nutrient Data Laboratory (NDL) web site since 1992. SR27 includes composition data for all the food groups and nutrients published in the 21 volumes of "Agriculture Handbook 8" (US Department of Agriculture 1976-92), and its four supplements (US Department of Agriculture 1990-93), which superseded the 1963 edition (Watt and Merrill, 1963). SR27 supersedes all previous releases, including the printed versions, in the event of any differences.

In July 2001, when NDL converted to a new version of its Nutrient Databank System (NDBS), formats were changed and fields added to improve the descriptive information for food items and the statistical information about the nutrient values. While data in previous releases have been moved to the new NDBS, they may not have been updated through the complete system. Therefore, many of these new fields contain data only for those items that have been processed through the new NDBS and it will take a number of years before they are populated for all food items in the database.

As part of this upgrade and in various modifications to the NDBS performed since then, various checks have been built into the system to ensure data integrity and quality control. Additional checks are performed after the SR files have been disseminated from the NDBS. These checks include making sure that, to the extent possible, all fields are complete. Another check is to make sure that various calculations are completed and correct, such as calculating carbohydrate by difference; calculating energy by multiplying protein, fat and carbohydrate by the appropriate factors; and calculating vitamin A from individual carotenoids. Other checks include making sure that values for related nutrients are complete. For example, if there are individual fatty acids, there should also be values for total saturated fatty acids, total monounsaturated fatty acids, and total polyunsaturated fatty acids. With some adaptation, the procedures described in FAO/INFOODS Guidelines for Checking Food Composition Data prior to the Publication of a User Table / Database (FAO/INFOODS, 2012) and Ahuja and Perloff (2008) have been used. Nutrient values are also compared from release to release to make sure any changes in the values can be explained. Reasons for these changes include new data generated by the National Food and Nutrient Analysis Program (NFNAP) analyses or provided by manufacturers, reformulations by the manufacturer, better food sampling, and changes in weighting based on market share data. Quality control procedures associated with the analytical data are described in the discussion of NFNAP under Notes on Foods (p. 54).

Data have been compiled from published and unpublished sources. Published data sources include the scientific literature. Unpublished data include those obtained from the food industry, other government agencies, and research conducted under contracts initiated by USDA's Agricultural Research Service (ARS). These contract analyses are currently conducted under NFNAP, in cooperation with the National Cancer Institute (NCI) and other offices and institutes of the National Institutes of Health (Haytowitz *et al*., 2008), the Centers for Disease Control and Prevention, and the Food and Drug Administration. Data from the food industry represents the nutrient content of a specific food or food product at the time the data is sent to NDL. The values may change due to reformulations or other processing changes by individual companies between the time that SR is released and the next update of SR. Values in the database may be based on the results of laboratory analyses or calculated by using appropriate algorithms, factors, or recipes, as indicated by the source code in the Nutrient Data file. Every food item does not contain all of the nutrients/components released in SR.

Notes on Foods has been included in the documentation, starting with SR23 (2010), and has been placed after the references. When the earlier paper copies of Agriculture Handbook No. 8, Composition of Foods: Raw, Processed, Prepared were released in separate sections by food group, each contained a section called Notes on Foods. The Notes gave additional information about the foods, such as the definitions of lean and fat for meats or enrichment for grain products. For some food groups, a brief description of research projects conducted to generate nutrient data were described. For those food groups, where Notes on Foods are not included herein, the original versions are available in the printed "Agriculture Handbook 8" sections (US Department of Agriculture, 1976-92).

Specific Changes for SR27

A number of studies either conducted through NFNAP or by collaboration with various groups resulted in various changes to the database. The major changes to the database since the last release are listed below.

- Nutrient profiles were added for new foods and existing nutrient profiles were updated for SR27 using data generated by USDA through the NFNAP, submitted by the food industry, or using other publically available data. A complete list of the added food items can be found in the ADD_FOOD file and the updated nutrients in the CHG_NUTR file. The formats of these files can be found on p. 41.

  A major focus of this effort is to monitor those foods which are major contributors of sodium to the diet. The Food Surveys Research Group (FSRG), in collaboration with NDL, identified a group of select foods, termed 'Sentinel Foods', to be monitored as primary indicators for assessing change in the sodium content of foods in US food supply. In many cases, to reduce the sodium content of a food, the product must be reformulated using new ingredients or a different balance of ingredients, potentially affecting concentrations of nutrients other than sodium. Examples are fat, sugars, fiber, potassium, and fatty acids. Concentrations of

these nutrients are also being monitored. The results of these analyses have been used to update nutrient profiles for foods in SR. An additional 1,300 foods, mainly commercially packaged and restaurant foods with added sodium used in the national dietary survey What We Eat in America (WWEIA), National Health and Nutrition Examination Survey (NHANES) are being monitored through information obtained directly from manufacturers or restaurant chains and their websites or changes in the Nutrition Facts Panel values. If there was an appreciable change in the sodium value, or any of the other aforementioned nutrients, SR was updated as needed.

- Foods added or updated include: Breakfast cereals, fried chicken pieces (breast, thigh, and wing) and skin and breading; fast food biscuit; chicken strips; hash browns; chicken noodle condensed soup; fried shrimp (from restaurant); several pulses (chickpeas, green and red lentils, and green peas); a number of vegetarian items; deli-style roast beef; enhanced and non-enhanced pork loin chops; rotisserie chicken breast; Italian-style meatballs; turkey bacon; popular juice smoothies and fortified juice products; greek yogurt; sorghum grain and flour; green tea; energy drinks and other beverages.

- A number of pizzas, served at school lunches, have been added.

- Foods from Mexican and Italian casual dining and full service restaurants were added. Added Mexican restaurant items include: cheese enchilada, cheese quesadilla, cheese tamales, soft beef tacos, refried beans and spanish rice. Added Italian restaurant items include: cheese ravioli, chicken parmesan, lasagna, spaghetti with meat sauce, and spaghetti with pomodoro sauce.

- A study was conducted to update nutrient data on ground beef, reflecting current market trends. For more details see Notes on Foods for Beef Products (p. 66).

- A study was conducted in collaboration with Colorado State University (CSU) to obtain nutrient information for 7 representative retail veal cuts. For more information, see Notes on Foods for Lamb, Veal, and Game Products (p. 85).

- In collaboration with the New Zealand Meat Industry Association, a study was undertaken to determine nutrient composition of imported New Zealand beef and lamb cuts and offal items which are available in retail grocery stores in the US. For more details, see Notes on Foods for Beef Products (p. 72), and for Lamb, Veal, and Game (p. 85).

- A study was undertaken in collaboration with Cornell University to determine the nutrient content of ruffed grouse and Canada goose. For more information see Notes on Foods for Poultry Products (p. 122).

- A Glossary of Cooking Methods for Meat and Poultry Items was created, to provide definitions for methods used in USDA studies when cooking meats and poultry prior to nutrient analysis. The Glossary is available in Appendix C.

- Products such as mixed dishes, soups and breakfast cereals which are no longer on the market or without current data have been removed. A complete list of deleted food items can be found in the DEL_FOOD file. The format of the file is given on p. 42.

- Notes on Foods for Beef Products (Food Group 13, p. 62), Pork Products (Food Group 10, p. 104) and Poultry Products (Food Group 05, p. 113) have been updated to include new information. With SR27, Notes on Foods for Lamb, Veal and Game Products (p.85) Have been added.

Data Files

The data files for SR27 are available in ASCII (ISO/IEC 8859-1) format and as a Microsoft Access 2007 database. A description of each field in these files and the relationships between each begins on p. 27. The Access database contains all the SR27 files and relationships, with a few sample queries and reports. An abbreviated file (p. 38) with fewer nutrients (46) but all the food items is also included. A Microsoft Excel 2007 spreadsheet of this file is also provided. These database and spreadsheet files are generally compatible with later releases of the same software package or with other software packages released at the same time.

Database Reports

The data in SR27 are available as page images containing all the data for each food. These data are separated into files by food groups. Adobe Reader is needed to see these files. There is a link from the NDL web site to Adobe's web site where it can be downloaded at no charge. Previously, reports containing selected foods and nutrients sorted either alphabetically or by nutrient content per household measure were available as PDF files. Starting with SR26, these reports are no longer available as the online search (http://ndb.nal.usda.gov) has been upgraded to include the ability to allow users to generate their own custom reports by selecting "Nutrient List" from the list of options. Users can select up to three nutrients from the database and generate reports for either all foods in SR27 or an abridged list (a shorter list of about 1,000 foods adapted from those in our publication: "US Department of Agriculture Home and Garden Bulletin 72, Nutritive Value of Foods" (Gebhardt and Thomas, 2002)). The user can also limit the report to a single food group or several food groups of their choosing. The reports can also be presented per 100 grams or per common household measure. The reports can be saved as either a PDF report or as a comma-delimited text file (csv), which can be opened in Excel or used with other programs.